

FAKRUL ISLAM TUSHAR

CURRICULUM VITAE

Work Address:

Department of Radiology and Imaging Sciences
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Education

- Ph.D. 2025 **Duke University**, NC, USA
Electrical and Computer Engineering (ML track).
Dissertation: *“Bridging the Gap: Integrating Clinical Knowledge, Simulation and Generative AI for Lung Cancer Diagnosis”*
Advisor: Joseph Y. Lo, PhD.
- M.Sc. 2019 **University of Girona**, Catalonia, Spain
Medical Imaging and Applications (MAIA), Erasmus Mundus Joint Master.
Thesis: *“Weakly-supervised multi-organ multi-disease classification using CT”*.
- B.Sc. 2017 **American International University-Bangladesh (AIUB)**, Bangladesh
Electrical and Electronics Engineering. *Cum Laude* and Dean’s Award.
Thesis: *“Development of a Modular Biopotential Amplifier Trainer for Biomedical Instrumentation Laboratory Experiments”*.

A. Positions, Honors and Service

- Positions and Employment

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|--------------------------|----------------|---|
| • Assistant Professor | 2026 – Present | Dept. of Radiology and Imaging Sciences, University of Arizona. Tucson, AZ. |
| • Principal Investigator | 2026 – Present | Tushar Lab , University of Arizona. Tucson, AZ. |
| • Research Associate | 2019 – 2021 | Duke University Medical Center. Durham, NC. |

- Honors and Awards

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|------------------------------------|-----------|---|
| • Best Poster Presentation | 2024 | International Summit of Virtual Imaging Trials in Medicine. |
| • Conference Travel Grant | 2024 | SPIE Medical Imaging Conference. |
| • Best Poster Award | 2022 | Pratt School of Engineering, Duke University. |
| • Master Thesis Grant | 2018 | Duke University Medical Center. |
| • Erasmus Mundus Scholarship | 2017–2019 | European Commission (€42,000). |
| • Academic Honor, <i>Cum Laude</i> | 2017 | AIUB. |
| • Dean’s Award | 2016 | Undergraduate final project ranked 2nd of 180, AIUB. |
| • Merit Scholarship | 2013–2017 | AIUB. |

- Scientific Reviewing

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| • Reviewer | 2021 – Present | <i>Artificial Intelligence in Medicine</i> |
| • Reviewer | 2024 – Present | <i>Scientific Reports</i> |
| • Reviewer | 2025 – Present | <i>Radiology: Artificial Intelligence</i> |

B. Publications

- Submitted/ In Review

1. 2026 **Tushar FI**, Lo JY. “Tri-Reader: An Open-Access, Multi-Stage AI Pipeline for First-Pass Lung Nodule Annotation in Screening CT.” *arXiv*. DOI: [10.48550/arXiv.2601.19380](https://doi.org/10.48550/arXiv.2601.19380).
2. 2026 **Tushar FI**, Wang A, Dahal L, Samei E, Harowicz MR, Kalpathy-Cramer J, Lafata KJ, Tailor TD, Rudin C, Lo JY. “Reproducible Benchmarking for Lung Nodule Detection and Malignancy Classification Across Multiple Low-Dose CT Datasets.” *arXiv*. DOI: [10.48550/arXiv.2405.04605](https://doi.org/10.48550/arXiv.2405.04605).
3. 2025 **Tushar FI**, Samei E, Rudin C, Lo JY. “NodMAISI: Nodule-Oriented Medical AI for Synthetic Imaging.” *arXiv*. DOI: [10.48550/arXiv.2512.18038](https://doi.org/10.48550/arXiv.2512.18038).
4. 2025 **Tushar FI**, Dahal L, McCabe C, Ho FC, Segars P, Abadi E, Lafata KJ, Samei E, Lo JY. “SYN-LUNGS: Towards Simulating Lung Nodules with Anatomy-Informed Digital Twins for AI Training.” *arXiv*. DOI: [10.48550/arXiv.2502.21187](https://doi.org/10.48550/arXiv.2502.21187).
5. 2025 Garcia-Alcoser ME, Ghogh Nejad M, **Tushar FI**, Kim D, Lafata KJ, Rubin GD, Lo JY. “Evaluating Large Language Models for Zero-Shot Disease Labeling in CT Radiology Reports Across Organ Systems.” *arXiv*. DOI: [10.48550/arXiv.2506.03259](https://doi.org/10.48550/arXiv.2506.03259).

- Full Articles Published

1. 2026 **Tushar FI**, Dahal L, Sotoudeh-Paima S, Abadi E, Segars WP, Lo JY, Samei E. “Utility of the virtual imaging trials methodology for objective characterization of AI systems and training data.” *Journal of Medical Imaging* 13(1):014506–014506. DOI: [10.1117/1.JMI.13.1.014506](https://doi.org/10.1117/1.JMI.13.1.014506). PMID: [41700300](https://pubmed.ncbi.nlm.nih.gov/41700300/). Project page: ReviCOVID.github.io.
2. 2026 Ghosh D, **Tushar FI**, Dahal L, Vancoillie L, Lafata KJ, Samei E, Lo JY, Luo S. “Demographic distribution matching between real-world and virtual phantom population.” *Medical Physics* 53(3):e70364. DOI: [10.1002/mp.70364](https://doi.org/10.1002/mp.70364). PMID: [41746164](https://pubmed.ncbi.nlm.nih.gov/41746164/)
3. 2025 **Tushar FI**, Vancoillie L, McCabe C, Kavuri A, Dahal L, Harrawood BP, Fryling M, Zarei M, Sotoudeh-Paima S, Ho FC, Ghosh D, Harowicz MR, Tailor TD, Luo S, Segars WP, Abadi E, Lafata KJ, Lo JY, Samei E. “Virtual Lung Screening Trial (VLST): An in silico study inspired by the National Lung Screening Trial for lung cancer detection.” *Medical Image Analysis* 103:103576. DOI: [10.1016/j.media.2025.103576](https://doi.org/10.1016/j.media.2025.103576). PMID: [40209556](https://pubmed.ncbi.nlm.nih.gov/40209556/). Project page: fitushar.github.io/VLST.github.io/.
4. 2025 Wang AJ, **Tushar FI**, Harowicz MR, Tong BC, Lafata KJ, Tailor TD, Lo JY. “The Duke Lung Cancer Screening (DLCS) Dataset: A Reference Dataset of Annotated Low-dose Screening Thoracic CT.” *Radiology: Artificial Intelligence* 7(4):e240248. DOI: [10.1148/ryai.240248](https://doi.org/10.1148/ryai.240248). PMID: [40237601](https://pubmed.ncbi.nlm.nih.gov/40237601/).
5. 2025 Wang Y, Gupta A, **Tushar FI**, Riley B, Wang A, Tailor TD, Tantum SL, Liu J-G, Bashir MR, Lo JY, Lafata KJ. “Concordance-based Predictive Uncertainty (CPU)-Index: Proof-of-concept with application towards improved specificity of lung cancers on low dose screening CT.” *Artificial Intelligence in Medicine* 160:103055. DOI: [10.1016/j.artmed.2024.103055](https://doi.org/10.1016/j.artmed.2024.103055). PMID: [39721356](https://pubmed.ncbi.nlm.nih.gov/39721356/).
6. 2025 Dahal L, Ghogh Nejad M, Vancoillie L, Ghosh D, Bhandari Y, Kim D, Ho FC, **Tushar FI**, Luo S, Lafata KJ, Abadi E, Samei E, Lo JY, Segars WP. “XCAT 3.0: A comprehensive library of personalized digital twins derived from CT scans.” *Medical Image Analysis* 103:103636. DOI: [10.1016/j.media.2025.103636](https://doi.org/10.1016/j.media.2025.103636). PMID: [40347915](https://pubmed.ncbi.nlm.nih.gov/40347915/).
7. 2025 Xia SJ, Vancoillie L, Sotoudeh-Paima S, Zarei M, Ho FC, **Tushar FI**, Chen X, Dahal L, Lafata KJ, Abadi E, et al. “Evaluation of unified harmonization of CT images across multiple tasks: A step towards AI generalizability.” *Medical Physics* 52(11):e70084. DOI: [10.1002/mp.70084](https://doi.org/10.1002/mp.70084). PMID: [41186937](https://pubmed.ncbi.nlm.nih.gov/41186937/)
8. 2022 **Tushar FI**, D’Anniballe VM, Hou R, Mazurowski MA, Fu W, Samei E, Rubin GD, Lo JY. “Classification of Multiple Diseases on Body CT Scans Using Weakly Supervised Deep Learning.”

Radiology: Artificial Intelligence 4(1):e210026. DOI: [10.1148/ryai.210026](https://doi.org/10.1148/ryai.210026). PMID: [35146433](https://pubmed.ncbi.nlm.nih.gov/35146433/). Code: github.com/fitushar/multi-label-weakly-supervised-classification-of-body-ct.

9. 2022 D’Anniballe VM*, **Tushar FI***, Faryna K, Han S, Mazurowski MA, Rubin GD, Lo JY. “Multi-label annotation of text reports from computed tomography of the chest, abdomen, and pelvis using deep learning.” *BMC Medical Informatics and Decision Making* 22(1):102. DOI: [10.1186/s12911-022-01843-4](https://doi.org/10.1186/s12911-022-01843-4). PMID: [35428335](https://pubmed.ncbi.nlm.nih.gov/35428335/). Code: github.com/fitushar/multi-label-annotation-text-reports-body-CT. *Joint First Author.
10. 2020 Hasan MK, Dahal L, Samarakoon PN, **Tushar FI**, Marti R. “DSNet: Automatic dermoscopic skin lesion segmentation.” *Computers in Biology and Medicine* 120:103738. PMID: [32421644](https://pubmed.ncbi.nlm.nih.gov/32421644/). DOI: [10.1016/j.compbiomed.2020.103738](https://doi.org/10.1016/j.compbiomed.2020.103738).

- Conference Proceedings

1. 2025 Garcia-Alcoser ME, **Tushar FI**, Ghogh Nejad M, Rubin GD, Lo JY. “Multidisease Classification of CT Reports Using Traditional Natural Language Processing and a Lightweight Foundation Model.” *Proc. SPIE 13411, Medical Imaging 2025: Imaging Informatics*, 61–67. DOI: [10.1117/12.3047690](https://doi.org/10.1117/12.3047690).
2. 2024 **Tushar FI**, Vancoillie L, McCabe C, Kavuri A, Dahal L, Harrawood B, Fryling M, Zarei M, Sotoudeh-Paima S, Ho FC, Ghosh D, Luo S, Segars WP, Abadi E, Lafata KJ, Samei E, Lo JY. “Virtual NLST: Towards Replicating National Lung Screening Trial.” *Proc. SPIE 12925, Medical Imaging 2024: Physics of Medical Imaging*, 129252F. DOI: [10.1117/12.3006915](https://doi.org/10.1117/12.3006915).
3. 2022 **Tushar FI**, Abadi E, Sotoudeh-Paima S, Fricks RB, Mazurowski MA, Segars WP, Samei E, Lo JY. “Virtual vs. Reality: External Validation of COVID-19 Classifiers Using XCAT Phantoms for Chest Computed Tomography.” *Proc. SPIE 12033, Medical Imaging 2022: Computer-Aided Diagnosis*, 1203305. DOI: [10.1117/12.2613010](https://doi.org/10.1117/12.2613010).
4. 2022 **Tushar FI**, Nujaim H, Fu W, Abadi E, Mazurowski MA, Segars WP, Samei E, Lo JY. “Quality or Quantity: Toward a Unified Approach for Multi-Organ Segmentation in Body CT.” *Proc. SPIE 12031, Medical Imaging 2022: Physics of Medical Imaging*, 679–684. DOI: [10.1117/12.2613101](https://doi.org/10.1117/12.2613101).
5. 2022 **Tushar FI**, D’Anniballe VM, Rubin GD, Samei E, Lo JY. “Co-occurring Diseases Heavily Influence the Performance of Weakly Supervised Learning Models for Classification of Chest CT.” *Proc. SPIE 12033, Medical Imaging 2022: Computer-Aided Diagnosis*, 675–679. DOI: [10.1117/12.2612700](https://doi.org/10.1117/12.2612700).
6. 2020 Saha A*, **Tushar FI***, Faryna K, D’Anniballe VM, Hou R, Mazurowski MA, Rubin GD, Lo JY. “Weakly Supervised 3D Classification of Chest CT Using Aggregated Multi-Resolution Deep Segmentation Features.” *Proc. SPIE 11314, Medical Imaging 2020: Computer-Aided Diagnosis*, 39–44. DOI: [10.1117/12.2550857](https://doi.org/10.1117/12.2550857). *Joint co-first author.
7. 2019 **Tushar FI**, Alyafi B, Hasan MK, Dahal L. “Brain Tissue Segmentation Using NeuroNet with Different Pre-Processing Techniques.” *2019 Joint 8th International Conference on Informatics, Electronics & Vision (ICIEV) and 2019 3rd International Conference on Imaging, Vision & Pattern Recognition (icIVPR)*, 223–227. IEEE. DOI: [10.1109/ICIEV.2019.8858515](https://doi.org/10.1109/ICIEV.2019.8858515).

- Abstracts/Posters

1. 2024 **Tushar FI**, Vancoillie L, Ghosh D, Lafata KJ, Samei E, Lo JY. “Virtual Human Twins in Lung Health: A Comprehensive In Silico Screening Approach.” RSNA Annual Meeting, Scientific Poster T5A-SPPH-2, Chicago, IL.
2. 2024 **Tushar FI**, Vancoillie L, Ghosh D, Lafata KJ, Lo JY. “Beyond Detection: Bridging the Gap Between Virtual Imaging Trials and Clinical Impact.” *Proc. Virtual Imaging Trials in Medicine 2024*, p. 202.

C. Invited Seminars / Workshops

- Invited Talks

1. 2024 “Unlocking the Power of AI: In Silico Trials in Chest Radiology.” CVIT Monthly Research Forum, Duke University. June 21.

- Podium Presentations at Conferences

1. 2026 “VLST-2: A Clinically Grounded In-Silico Lung Screening Cohort for Virtual Imaging Trial Design.” Virtual Imaging Trials in Medicine 2026 mini, Duke University. April 27.

D. Open-Source Contributions

- Tools & Codes

1. **PiNS:** Point-driven Nodule Segmentation github.com/fitushar/PiNS
2. **CaNA:** Context-Aware Augmentation github.com/fitushar/CaNA
3. **NoMAISI:** Nodule-Oriented AI for Synthetic Imaging github.com/fitushar/NoMAISI
4. **TriAnnot:** AI-Based Annotation Framework github.com/fitushar/TriAnnot
5. **AI in Lung Health Benchmarking** [GitHub](#)
6. **Radiology NLP:** Weakly-supervised Classification [GitHub](#)
7. **3D Medical Imaging Pre-Processing** [GitHub](#)

- Data Resources

1. **DLCS24:** Duke Lung Cancer Screening Dataset [zenodo.13799069](https://zenodo.org/record/13799069)
2. **NLST-3D:** 3D Annotated Lung Subset [zenodo.15320923](https://zenodo.org/record/15320923)
3. **U-10 COVID-19 CTs:** Multi-site Dataset [zenodo.14064172](https://zenodo.org/record/14064172)
4. **HAID:** Health AI Data Resource github.com/fitushar/HAID

E. Teaching

- Duke University

Course	Term	Enrollment	Role
ECE 580: Intro to Machine Learning	Spring 2025	40+ (Grad)	Teaching Assistant
ECE 653: Human-Centered Computing	Spring 2023	40+ (Grad)	Teaching Assistant

F. Student Mentorship

- Research Mentorship

Student	Term	Level	Project
Muhammad Zain	Spring 2024	Graduate	XAI for lung lesions
Enrique Almar-Munoz	Spring 2023	Graduate	Active learning for lung lesions
Husam Nujaim	Summer 2022	Intern	CT segmentation
Anindo Saha	Summer 2020	Intern	Imaging feature aggregation
Khrystyna Faryna	Summer 2020	Intern	Radiology text classification

References

- **Geoffrey Rubin, MD** — Professor and Chair, Department of Radiology and Imaging Sciences, University of Arizona.
grubin@arizona.edu
- **Joseph Y. Lo, PhD** — Professor and Vice Chair for Research of Radiology, Duke University.
joseph.lo@duke.edu
- **Ehsan Samei, PhD** — Professor of Radiology, Physics, BME, and ECE, Duke University.
esi.samei@duke.edu